GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
SOFTWARE ENGINEERING

Course Code: GR22A3052 **L/T/P/C: 3/0/0/3**
III Year I Semester

Course Outcomes:

1. Understand business requirements and choose a relevant Process model for a given software proposal
2. Analyze the requirements to prepare SRS.
3. Model various Architectural Designs for a software project.
4. Develop various functional and structural test cases for a software module
5. Estimate the Cost and Schedules of a Software Project.

UNIT I

Introduction to Software Engineering: The evolving role of software, Changing Nature of Software, Software myths.

A Generic view of process: Software engineering- A layered technology, a process framework, The Capability Maturity Model Integration (CMMI), Process patterns, process assessment, personal and team process models.

Process models: The waterfall model, Incremental process models, Evolutionary process models, The Unified process.

UNIT II

Software Requirements: Functional and non-functional requirements, User requirements, System requirements, Interface specification, the software requirements document.

Requirements engineering process: Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management.

UNIT III

Design Engineering: Design process and Design quality, Design concepts, the Design model.

Creating an architectural design: Software architecture, Data design, Architectural Styles and Patterns, Architectural Design.

Performing User interface design: Golden rules, User interface analysis and design, interface analysis, interface design steps, Design evaluation.

UNIT IV

Testing Strategies: A strategic approach to software testing, test strategies for conventional software, Black-Box and White-Box testing, Validation testing, System testing, the art of Debugging.

Product metrics: Software Quality, Metrics for Analysis Model, Metrics for Design Model, Metrics for source code, Metrics for testing, Metrics for maintenance.

Metrics for Process and Products: Software Measurement, Metrics for software quality.

UNIT V

Risk management: Reactive vs. Proactive Risk strategies, software risks, Risk identification, Risk projection, Risk refinement, RMMM, RMMM Plan.

Quality Management : Quality concepts, Software quality assurance, Software Reviews, Formal technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality standards.

Text Books

1. Software Engineering, A practitioner's Approach- Roger S.Pressman, 6th edition. McGrawHill International Edition.
2. Software Engineering- Sommerville, 7th edition, Pearson education.

References

1. Software Engineering- K.K. Agarwal & Yogesh Singh, New Age International Publishers
2. Software Engineering, an Engineering approach- James F. Peters, Witold Pedrycz, John Wiely.
3. Systems Analysis and Design- Shely Cashman Rosenblatt,Thomson Publications.
4. Software Engineering principles and practice- Waman S Jawadekar, The McGraw-Hill Companies.

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
ARTIFICIAL INTELLIGENCE

Course Code: GR22A3070 **L/T/P/C: 3/0/0/3**
III Year I Semester

Prerequisites:

A Course in Artificial intelligence would require the knowledge of following concepts

- Logic Theory
- Probability Theory
- Numerical Analysis
- Operations on Matrices

Course Outcomes:

1. Select an appropriate searching strategy for developing intelligent agents to find solution in optimized way using building blocks of AI.
2. Apply Prepositional and First Order Logic methods to resolve decisions for Knowledge-based agents.
3. Practice uncertain knowledge and reasoning handling using Bayesian Networks.
4. Analyze the working of temporal model, Hidden Markov Model, and Decision trees.
5. Breakdown the steps to construct small robot capable of performing perception and movement based on techniques learnt in the course.

UNIT I

Introduction to AI: Introduction, Foundation of AI, History of Intelligent Agents, Agents and environments, Concept of Rationality, Nature of environments & Structure of Agents, Problem solving agents and formulation, Searching For Solutions and Strategies, Uninformed search strategies BFS, DFS, Heuristic approach, Greedy best search, A* Search, Game Playing: Adversal search, Games, Min-Max algorithm, Optimal decisions in multiplayer games, Alpha Beta pruning.

UNIT II

Knowledge Representation & Reasons: Logical agents, Knowledge based agents, The Wumpus world, Logic: Proportional logic, Resolution patterns in proportional logics, Resolution: Forward and Backward chaining, First order logic: Inference in First order logic, Proportional vs first order inference, Unification & Lifting, forward chaining, Resolution, Practice problems.

UNIT III

Uncertain Knowledge and Reasoning: Uncertainty-Acting under uncertainty, Basic probability notion, the axioms of probability, inference using full joint distribution, Independence, Bayes' rule.

Probabilistic Reasoning: Representing Knowledge in uncertain domain, the semantics of Bayesian networks, efficient representations of conditional distributions, exact inference in

Bayesian networks, approximate inference in Bayesian networks.

UNIT IV

Probabilistic reasoning over time: Time and uncertainty, inference in temporal model, Hidden Markov models.

Planning: Overview, Components of Planning, Conditional Planning, Goal Stack Planning.

Learning: Learning from observations: Forms of learning, inductive learning, learning decision trees, ensemble learning, why learning works.

UNIT V

Perception: Image formation, Early Image Processing operations- Edge detection, image segmentation. Object recognition, using vision for manipulation and navigation.

Robotics: Introduction, Robot hardware, robotic perception, planning to move, robotic software architectures, application domains.

Introduction to Expert Systems, Conversational AI, Generative AI.

Text Books:

1. Artificial Intelligence: A Modern Approach, Staurt Russel, Peter Norvig, 4th Edition, Pearson

References:

1. Artificial Intelligence, Elaine Riche & Kevin Night, 2nd Edition, Tata McGraw Hill Education Pvt Ltd.
2. Artificial Intelligence, Kevin Knight, Elaine Rich, Shivashankar B. Nair, 3rd Edition, McGraw Hill Education Pvt Ltd.
3. Paradigms of Artificial Intelligence Programming, Case Studies in Common Lisp, Peter Norvig, Morgan Kaufmann Publisher.

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
WEB PROGRAMMING

Course code: GR223053 **L/T/P/C: 3/0/0/3**
III Year I Semester

Prerequisites

1. Knowledge in core Programming

Course Outcomes

1. Develop web page using JavaScript for event handling which uses HTML tags and intrinsic event attributes.
2. Understand the concept and learn to use the building blocks of XML and Java Bean Components.
3. Build server side applications using servlets for web applications.
4. Design dynamic and interactive websites using JSP.
5. Design databases and Develop the supporting code for Client and Server-Side applications using JSP and Servlets

UNIT I

HTML: Common tags- List, Tables, images, forms, Frames, Cascading Style sheets.

Java Script: Introduction to Java Scripts, Objects in Java Script, Dynamic HTML with Java Script

UNIT II

XML: Introduction to XML, Building Blocks in XML, Document type definition, XML Schemas, Presenting XML.

Java Beans: Introduction to Java Beans, Advantages of Java Beans, Java Beans API and Features of Java Beans: Introspection, Bound properties, Constrained properties, Persistence, Customization.

UNIT III

Servlets: Introduction, Lifecycle, Generic Servlet Package, reading parameters, Reading Initialization parameters, HTTP Servlet Package, Handling Http Request & Responses, Cookies, Session Tracking.

UNIT - IV

JSP Application Development: The Problem with Servlet, The Anatomy of a JSP Page, JSP Processing, JSP Components: Directives, Action Elements, Scripting Elements, Tag Libraries, Expression Language, Java Bean Components, Deploying JAVA Beans in a JSP Page, Model View Controller, JSP Application Design with MVC Setting Up, Error Handling, Scope of Implicit Objects.

UNIT V

Database Access: Database Programming using JDBC, Studying javax.sql.* package, Steps to access database, Working with Prepared Statements, Accessing a Database from a

Java/Servlet/JSP, Application – Specific Database Actions, Introduction to struts framework.

TEXTBOOKS:

1. Web Programming, building internet applications, Chris Bates 2nd edition, WILEY, Dreamtech
2. The complete Reference Java 2 Fifth Edition by Patrick Naughton and Herbert Schildt. TMH
3. Java Server Pages –Hans Bergsten, SPD O'Reilly

REFERENCE BOOKS:

1. Programming world wide web-Sebesta, Pearson
2. Core SERVLETS ANDJAVASERVER PAGES VOLUME 1: CORE TECHNOLOGIES By Marty Hall and Larry Brown Pearson
3. Internet and World Wide Web – How to program by Dietel and Nieto PHI/Pearson Education Asia.
4. Jakarta Struts Cookbook , Bill Siggelkow, S P D O'Reilly for chap 8.
5. Murach's beginning JAVA JDK 5, Murach, SPD

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
COMPUTER GRAPHICS
(PROFESSIONAL ELECTIVE – I)

Course code:GR22A3054 **L/T/P/C: 3/0/0/3**
III Year I Semester

Prerequisites:
Student should have knowledge of the following mathematical concepts:
1. Matrices
2. Basic linear algebra such as solving a system of linear equations
3. Polynomials
4. Elementary signal processing (Fourier transform and filtering)

Course Outcomes:
1. Summarise the basic concepts used in computer graphics.
2. Implement various algorithms to draw line, circle, scan and convert the basic geometrical primitives.
3. Apply different algorithms for drawing 2D primitives such as transformations, area filling and clipping.
4. Analyse the importance of 3D modelling and projections.
5. Demonstrate the fundamentals of animation, virtual reality and its related technologies.

UNIT I
Introduction to computer graphics- Introduction, Non interactive/interactive Graphics, Uses of computer graphics, classification of Applications, Programming Language, Graphics system configuration

Graphic Systems- Introduction, Cathode Ray Tube(CRT)basics, Refresh Display, Raster Display, Computer Graphic Software, Integration of Graphics Standard

UNIT II
Output Primitives- Introduction, Representing Image, Straight Line, Line drawing algorithms, DDA Algorithm, Bresenham's Line Algorithm, Circle generating Algorithm, Bresenham's circle Algorithm, Midpoint circle Algorithm, Polygon filling Algorithms, Character or Text Generation, Aliasing and Antialiasing.

UNIT III
Two Dimensional Transformations-Introduction, Representation of points, Matrix Algebra and Transformation, Transformation of points, Transformation of straight line, Midpoint Transformation, Transformation of Parallel Lines, Transformation of Intersecting Lines, Rotation

Window Clipping- Introduction, Viewing Transformation, Clipping, Point Clipping, Line Clipping, Cohen-Sutherland Line clipping, Polygon Clipping, Sutherland-Hodgman Algorithm, Curve Clipping

UNIT IV

3D Concepts and Techniques- Introduction, 3D Transformations, Rotation about an axis Parallel to a Coordinate Axis, Rotation about an Arbitrary Axis in Space, Reflection through an Arbitrary Plane, 3D Modeling Schemes, Projection, Orthographic Projection, Isometric Projection, Oblique Projection, perspective projection

UNIT V

Introduction to Multimedia- Pc specification, visual elements, wav and mp3 format, sound elements, multimedia storage, flash animation.

TEXTBOOKS:

1. Computer Graphics, Amarendra N Sinha, Arun DUdai, TataMcGrawHill
2. Fundamentals of Multimedia, Ze-Nian Li, Mark S. Drew, Pearson Prentice Hall

REFERENCE BOOKS

1. Multimedia and communications technology, Steve Heath,Elsevier
2. Mathematical Elements for Computer Graphics,2nd Edition, David F. Rogers, J.AlanAdams

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
PRINCIPLES OF PROGRAMMING LANGUAGES
(PROFESSIONAL ELECTIVE – I)

Course Code:GR22A3047 **L/T/P/C: 3/0/0/3**
III Year I Semester

Course Outcomes:
1. Evaluate language constructs and programming paradigms.
2. Apply formal methods of syntax, semantics and data types.
3. Illustrate sub programs, blocks and control structures in different programming languages.
4. Construct abstract data types, concurrency and exceptions.
5. Compare functional and imperative languages.

UNIT I
Preliminary Concepts: Reasons for studying, concepts of programming languages, Programming domains, Language Evaluation Criteria, Influences on Language design, Language categories, Programming Paradigms – Imperative, Object Oriented, Functional Programming , Logic Programming.
Programming Language Implementation: Compilation and Virtual Machines, Programming environments.

UNIT II
Syntax and Semantics: General Problem of describing Syntax and Semantics, formal methods of describing syntax - BNF, EBNF for common programming language features, parse trees, ambiguous grammars, attribute grammars, denotation semantics and axiomatic semantics for common programming language features.
Data types: Introduction, primitive, character, user defined, array, associative, record, union, pointer and reference types, design and implementation uses related to these types, Names, Variable, concept of binding, type checking, strong typing, type compatibility, named constants and variable initialization.

UNIT III
Expressions and Statements: Arithmetic relational and Boolean expressions, Short circuit evaluation, mixed mode assignment, Assignment Statements, Control Structures– Statement Level, Compound Statements, Selection, Iteration, Unconditional Statements, guarded commands.
Subprograms and Blocks: Fundamentals of sub-programs, Scope and lifetime of variable, static and dynamic scope, Design issues of subprograms and operations, local referencing environments, parameter passing methods, overloaded sub-programs, generic sub-programs, parameters that are sub- program names, design issues for functions, user defined overloaded operators, co routines.

UNIT IV

Abstract Data types: Abstractions and encapsulation, Introduction to data abstraction, design issues, language examples, C++ parameterized ADT, object oriented programming in C++, Java, C#,Python

Concurrency: Subprogram level concurrency, semaphores, monitors, message passing, Java threads, Examples: Java RMI, Parallel Java, Parallel C

Exception handling: Exceptions, Exception propagation, Exception handler in C++ and Java and PHP. Handling Large Database.

UNIT V

Functional Programming Languages: Introduction, fundamentals of FPL, LISP, ML, Haskell, application of Functional Programming Languages and comparison of functional and imperative languages.

Logic Programming Language: Introduction and overview of logic programming, basic elements of prolog, application of logic programming.

TEXT BOOKS:

1. Concepts of Programming Languages Robert .W. Sebesta 6/e, Pearson Education.
2. Programming Languages –Louden, Second Edition,Thomson.

REFERENCES:

1. Programming languages –Ghezzi, 3/e, JohnWiley
2. Programming Languages Design and Implementation – Pratt and Zelkowitz, Fourth Edition PHI/Pearson Education

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
DATA WAREHOUSING AND DATA MINING
(PROFESSIONAL ELECTIVE – I)

Course Code:GR223069 **L/T/P/C: 3/0/0/3**
III Year I Semester

Prerequisites:
1. Students are expected to have knowledge of transactional and relational databases, probability and statistics.

Course Outcomes:
1. Demonstrate the concepts of data mining, its applications, and various pre-processing methods.
2. Develop a prototype for data warehouse and data mart for any organization.
3. Apply different types of association rule mining techniques for solving practical problems.
4. Implement classification models and algorithms in data warehouses and data mining.
5. Analyze clustering techniques and recent trends in advanced data mining techniques.

UNIT I
Introduction: Why Data mining, What is Data Mining, What Kinds of Data Can Be Mined, What Kinds of Patterns Can Be Mined, Which Technologies are used, Data Mining Task Primitives, Integration of a Data Mining System witha Database or a Data Warehouse System, Major issues in Data Mining.
Data Preprocessing: Need for Preprocessing the Data, Data Cleaning, Data Integration and Transformation, Data Reduction.

UNIT II
Data Warehouse and Online Analytical Processing: Data Warehouse-Basic concepts, Data Warehouse Modeling - Multidimensional Data Model, Star, Snowflake, Fact Constellation Schemas, Concept Hierarchies, Measures, Typical OLAP Operations, From Data Warehousing to Data Mining.
Data Warehouse Implementation - Efficient Data Cube Computation: Overview, Indexing OLAP Data – Bitmap Index, Join Index.

UNIT III
Data Characterization and Discrimination: Data Generalization by Attribute-Oriented Induction.
Mining Frequent Patterns, Associations and Correlations: Basic Concepts, Market Basket Analysis, Efficient and Scalable Frequent Item set Mining Methods, Mining various kinds of Association Rules, From Association Mining to Correlation Analysis, Constraint-Based Association Mining.

UNIT IV
Classification: Issues Regarding Classification and Prediction, Classification by Decision Tree Induction, Bayesian Classification, Rule-Based Classification, Classification by Back propagation, k-Nearest-Neighbor Classifiers.
Prediction: Introduction to Regression techniques, Linear Regression.

Accuracy and Error measures: Classifier Error measures, Predictor Error Measures, Evaluating the accuracy of Classifier and Predictor.

UNIT V

Cluster Analysis: What is Cluster Analysis, Types of Data in Cluster Analysis, Categorization of Major Clustering Methods, Partitioning Methods - k-Means, k-Mediods, Hierarchical Methods - AGNES and DIANA, Density-Based Methods - DBSCAN. Introduction to Mining Complex Data Types: Mining Data Streams, Mining Time- Series Data, Mining Sequence Patterns in Transactional Databases, Mining Complex Data Objects, Spatial Data Mining, Multimedia Data Mining, Text Mining, Mining the World Wide Web.

Teaching Methodologies:

- Power Point Presentations
- Tutorial Sheets
- Assignments

Textbooks:

1. Data Mining Concepts and Techniques - Jiawei Han & Micheline Kamber, Morgan Kaufmann Publishers, Elsevier, Third Edition, 2012.
2. Introduction to Data Mining – Pang-Ning Tan, Michael Steinbach and Vipin Kumar, Pearson education.

References:

1. Data Mining Techniques – Arun K. Pujari, Second Edition, Universities Press.
2. Data Warehousing in the Real World, Sam Aanhory and Dennis Murray, Pearson Edn Asia.

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
CYBER SECURITY
(PROFESSIONAL ELECTIVE – I)

Course Code:GR223055 **L/T/P/C: 3/0/0/3**
III Year I Semester

Prerequisites:

1. Basic communication methods.
2. Knowledge about cybercrimes.
3. Security primitives.

Course Outcomes:

1. Obtain firm understanding on basic terminology and concepts of cybercrimes and security.
2. Analyze the plans of attacks.
3. Deal with the security challenges posed by mobile devices.
4. Implement the tools to handle security challenges.
5. Evaluate the associated challenges and the cost of cybercrimes in Organizations.

UNIT I

Introduction to Cybercrime: Introduction, Cybercrime and Information Security, Cyber Criminals, Classifications of Cybercrimes and Cybercrime: The legal Perspectives and Indian Perspective, Cybercrime and the Indian ITA 2000, A Global Perspective on Cybercrimes.

UNIT II

Cyber Offenses: Introduction, How Criminals plan the Attacks, Types of Attackers, Cyber stalking, Cyber Cafe and Cybercrimes, Botnets: The Fuel for Cybercrime, Attack Vector, Cloud Computing.

UNIT III

Cybercrime: Mobile and Wireless Devices: Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication service Security, Attacks on Mobile/Cell Phones, Mobile Devices: Security Implications for Organizations, Organizational Measures for Handling Mobile, Organizational Security Policies and Measures in Mobile Computing Era, Laptops.

UNIT IV

Tools and Methods Used in Cybercrime: Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Key loggers and Spywares, Virus and Worms, Trojan Horse and Backdoors, Steganography, DoS and Types of DDoS attacks, SQL Injection, Buffer Overflow.

UNIT V

Cyber Security: Organizational Implications Introduction, Cost of Cybercrimes and IPR issues, Web threats for Organizations, Security and Privacy Implications, Social media marketing: Security Risks and Perils for Organizations, Social Computing, and the associated challenges for Organizations.

TEXTBOOKS:

1. Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, Nina Godbole and Sunil Belapure, WileyINDIA.

REFERENCES:

1. Cyber Security Essentials, James Graham, Richard Howard and Ryan Otson, CRCPress.
2. Introduction to Cyber Security, Chwan-Hwa(john) Wu,J.DavidIrwin.CRC Pr